

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2023
Math 2123
 (Mathematics - III)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Find the constants a, b, c so that the directional derivative of $\phi = ax^2y + byz + cz^2x^3$ at $(1, 2, -1)$ has a maximum of magnitude 64 in a direction parallel to the z axis. (11)
- (b) A fluid motion is given by $\vec{v} = (y\sin z - \sin x)\vec{i} + (x\sin z + 2yz)\vec{j} + (xycos z + y^2)\vec{k}$. Is the motion irrotational? If so, find the corresponding velocity potential. (12)
- (c) Find the values of constants λ and μ so that the surfaces $\lambda x^2 - \mu yz = (\lambda + 2)x, 4x^2y + z^3 = 4$ intersect orthogonally at the point $(1, -1, 2)$. (12)
2. (a) If $\vec{A} = (y - 2x)\vec{i} + (3x + 2y)\vec{j}$, compute the circulation of $\vec{A}$ about a circle C in the xy plane with center at the origin and radius 2, if C is traversed in the positive direction. (11)
- (b) Evaluate $\iint_S \vec{A} \cdot \vec{n} \, ds$, where $\vec{A} = z\vec{i} + x\vec{j} - 3y^2z\vec{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$. (12)
- (c) Find the Laplace transform of the following functions: (12)
 - (i) $F(t) = \frac{e^{-t} \sin t}{t}$
 - (ii) $G(t) = te^{-3t} \sin 2t$
3. (a) Evaluate $\iiint_V \vec{V} \cdot \vec{F} \, dv$, where V is the closed region bounded by the planes $x = 3, y = 2, z = 1, x = 4, y = 3$ and $z = 2$ and $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$. (11)
- (b) Find the inverse Laplace transform of (12)
 - (i) $\frac{3s+1}{(s-1)(s^2+1)}$
 - (ii) $\frac{1}{\sqrt{2s+3}}$
- (c) State the convolution theorem. Use the convolution theorem to find $L^{-1}\left\{\frac{1}{s^2(s+1)^2}\right\}$. (12)
4. (a) (Using Heaviside expansion formula) find the inverse Laplace transform of (10)

$$\frac{s-1}{(s+3)(s^2+2s+2)}$$
- (b) Using Laplace transform solve $y'' - 7y' + 2y = t + 4$, given that $y(0) = 0, y'(0) = -1$ (15)
- (c) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the path C : the straight line joining $(0, 0, 0)$ to $(1, 0, 0)$; where $\vec{F} = 2x\vec{i} - z\vec{j} + x\vec{k}$. (10)

Section - B

(09)

5. (a) Define with example:

- (i) Skew-symmetric matrix
- (ii) Diagonal matrix
- (iii) Orthogonal matrix

- (b) If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find A^{-1} . Also express matrix A and its inverse as products of elementary matrices performing for row and column elementary operations. Additionally verify that the A^{-1} you find is correct. (16)

- (c) Define linear independence of vectors. Check whether the vectors $\vec{a} = \hat{i} + 3\hat{j} + 5\hat{k}$, $\vec{b} = 2\hat{i} + 5\hat{j} + 9\hat{k}$ and $\vec{c} = -3\hat{i} + 9\hat{j} + 3\hat{k}$ are linearly independent or not. If possible, express dependent vector as a linear combination independent vectors. (10)

6. (a) What do you mean of the rank of a matrix? Reduce the matrix A, to its row echelon form, row canonical, and from normal form, where (18)

$$\begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$$

Also determine its rank. Moreover, if possible, find A^{-1} as a by-product of the process to obtain its normal form.

- (b) What is the value of "K" for which the following system has: (i) an unique solution, (ii) many solution, and (iii) no solution? If possible, find the unique solution of the system using the Gauss elimination method (elementary row operations, where (17)

$$\begin{bmatrix} 2 & 3 & 12 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ k \end{pmatrix}$$

7. (a) Given $2x - y + 3z = 0$; $2x + 2y + z = 0$; $x - 4y + 7z = 0$. What is the value of 'T' so that the system of equations has: (i) trivial solution (ii) non-trivial solution? If possible find the non-trivial solution of the system of equations for same value of 'T'. (15)

- (b) Write down five properties/definitions for any eigensystem. Find the eigenvalues and corresponding eigenvectors of A, where $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Also find the eigenvalues and corresponding eigenvectors of the matrix $A - 4I$, where I is the identity matrix. (20)

8. (a) A space curve is defined by the position vector $\cos t \hat{i} + \sin t \hat{j} + \sqrt{3} \hat{k}$. Find the unit tangent vector $\hat{T}$, normal vector $\hat{N}$, and binormal vector $\hat{B}$. Calculate the radius of curvature, acceleration, and to tangent and normal components of acceleration. Additionally, determine the trihedral plan that contains the vectors $\hat{T}$ and $\hat{N}$ at $t = \frac{\pi}{2}$. (28)

- (b) A force $\vec{F} = 10t\hat{i} + 15t^2\hat{j}$ is acting on a moving particle with a mass of 5 units. Find its velocity such that the velocity at $t = 0$ is $5\hat{k}$. (07)